cp_6b

1.

Find the Group | 30/30

Easy

The numbers from **1** to **10** are divided into two groups, **Group A** and **Group B**.

Group A	Group B
1	2
3	4
5	6
7	8
9	10

Write a program that reads a number `N` and prints the group in which the given number `N` is present. The number `N` is always from **1** to **10**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing a string representing the group of the number `N` .

Constraints

The number `N` given as input is always greater than **0** and less than or equal to **10**.

Explanation

For example, if the given number is `N = 6` ,

- The total number of groups is **2**.
- The number that leaves **1** as the remainder when divided by **2** is in **Group A**.
- The number that leaves **0** as the remainder when divided by **2** is in **Group B**.
- When **6** is divided by **2** the remainder is **0**. So, **6** should be in **Group B**.

The output should be **Group B** as the remainder is **0** when **6** is divided by **2**.

Sample Input 1

```
6
```

Sample Output 1

```
Group B
```

code

```
number=int(input())
if number%2==1:
    print("Group A")
elif number%2==0:
    print("Group B")
```

2.

Find the Group - 2 | 35/35

Solved

Easy

The numbers from **1** to **30** are divided into six groups, **Group 1**, **Group 2**, **Group 3**, **Group 4**, **Group 5** and **Group 6**.

Group 1	Group 2	Group 3	Group 4	Group 5	Group 6
1	2	3	4	5	6
7	8	9	10	11	12
13	14	15	16	17	18
19	20	21	22	23	24
25	26	27	28	29	30

Write a program that reads a number `N` and prints the group in which the given number `N` is present. The number `N` is always from **1** to **30**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing a string representing the group of the number `N` .

Constraints

The number `N` given as input is always greater than **0** and less than or equal to **30**.

Explanation

For example, if the given number is `N = 29` ,

▶ Detailed Explanation to find the group

- When **29** is divided by **6** the remainder is **5**. So, **29** will be in **Group 5**.

The output should be **Group 5**.

Sample Input 1

```
29
```

Sample Output 1

```
Group 5
```

Sample Input 2

```
3
```

Sample Output 2

```
Group 3
```

code:

```
number=int(input())
if number%6==0:
    print("Group 6")
elif number%6==1:
    print("Group 1")
elif number%6==2:
```

```
    print("Group 2")
elif number%6==3:
    print("Group 3")
elif number%6==4:
    print("Group 4")
elif number%6==5:
    print("Group 5")
```

3.

Denominations | 35/35

Easy

Write a program that reads an amount `A` and prints the minimum number of **5** and **1** rupee notes required for the given amount.

Input

The input will be a single line containing an integer representing the amount `A` .

Output

The first line of output should be a string containing the required number of **5** rupee notes as shown in the sample output.

The second line of output should be a string containing the required number of **1** rupee notes as shown in the sample output.

Explanation

For example, if the given amount `A = 16` ,

- The number of **5** rupee notes required for **16** is **3**. (`16 / 5 = 3.2`)
  - The remaining amount is **1**. (`16 - (5 * 3) = 1`)
- The number of **1** rupee notes required for **1** is **1**. (`1 / 1 = 1`)

Number of **5** rupee notes ⇒ 5 × 3 = 15
Number of **1** rupee notes ⇒ 1 × 1 = 1

Total ⇒ 16

The output should be,

```
5:3
1:1
```

Sample Input 1

```
16
```

Sample Output 1

```
5:3
1:1
```

Sample Input 2

```
102
```

Sample Output 2

```
5:20
1:2
```

code

```
amount=int(input())
no_of_5s=int(amount/5)
```

```
no_of_1s=amount%5
print("5:"+str(no_of_5s))
print("1:"+str(no_of_1s))
```

4.

Denominations - 2 | 40/40

Write a program that reads an amount A and prints the minimum number of **100**, **50**, **10** and **1** rupee notes required for the given amount.

Input

The input will be a single line containing an integer representing the amount `A` .

Output

The first line of output should be a string containing the required number of **100** rupee notes as shown in the sample output.

The second line of output should be a string containing the required number of **50** rupee notes as shown in the sample output.

The third line of output should be a string containing the required number of **10** rupee notes as shown in the sample output.

The fourth line of output should be a string containing the required number of **1** rupee notes as shown in the sample output.

Explanation

For example, if the given amount `A = 893` ,

- The number of **100** rupee notes required for **893** is **8**. (`893 / 100 = 8`)
  - The remaining amount is **93**. (`893 - (100 * 8) = 93`)
- The number of **50** rupee notes required for **93** is **1**. (`93 / 50 = 1.86`)
  - The remaining amount is **43**. (`93 - (50 * 1) = 43`)
- The number of **10** rupee notes required for **43** is **4**. (`43 / 10 = 4.3`)
  - The remaining amount is **3**. (`43 - (10 * 4) = 3`)
- The number of **1** rupee notes required for **3** is **3**. (`3 / 1 = 3`)

Number of **100** rupee notes ⇒ 100 × 8 = 800
Number of **50** rupee notes ⇒ 50 x 1 = 50
Number of **10** rupee notes ⇒ 10 x 4 = 40
Number of **1** rupee notes ⇒ 1 x 3 = 3

Total ⇒ 893

The output should be,

```
100:8
50:1
10:4
1:3
```

Sample Input 1

```
893
```

Sample Output 1

```
100:8
```

code:

```
amount=int(input())
no_of_100s=int(amount/100)
no_of_50s=int((amount%100)/50)
no_of_10s=int(((amount%100)%50)/10)
no_of_1s=int(((amount%100)%50)%10)
print("100:"+str(no_of_100s))
print("50:"+str(no_of_50s))
print("10:"+str(no_of_10s))
print("1:"+str(no_of_1s))
```

5.

Denominations - 3 | 25/25

Medium

Write a program that reads an amount `A` and prints the minimum number of **500**, **50**, **10** and **1** rupee notes required for the given amount.

Input

The input will be a single line containing an integer representing the amount `A` .

Output

The output should be a single line containing a string that has the number of **500**, **50**, **10** and **1** rupee notes required for the given amount `A` .

Explanation

For example, if the given amount `A = 1543` ,

- The number of **500** rupee notes required for **1543** is **3**. (`1543 / 500 = 3.086`)
  - The remaining amount is **43**. (`1543 - (500 * 3) = 43`)
- The number of **50** rupee notes required for **43** is **0**. (`43 / 50 = 0.86`)
- The number of **10** rupee notes required for **43** is **4**. (`43 / 10 = 4.3`)
  - The remaining amount is **3**. (`1543 - (500 * 3) - (10 * 4) = 3`)
- The number of **1** rupee notes required for **3** is **3**. (`3 / 1 = 3`)

Number of **500** rupee notes ⇒ 500 × 3 = 1500
Number of **50** rupee notes ⇒ 50 × 0 = 0
Number of **10** rupee notes ⇒ 10 × 4 = 40
Number of **1** rupee notes ⇒ 1 × 3 = 3

Total ⇒ 1543

The output should be `500: 3 50: 0 10: 4 1: 3` .

Sample Input 1

```
1543
```

Sample Output 1

```
500: 3 50: 0 10: 4 1: 3
```

Sample Input 2

```
1259
```

Sample Output 2

```
500: 2 50: 5 10: 0 1: 9
```

code

```
amount=int(input())
no_of_500s=int(amount/500)
```

```
no_of_50s=int((amount%500)/50)
no_of_10s=int(((amount%500)%50)/10)
no_of_1s=int(((amount%500)%50)%10)
print("500: "+str(no_of_500s)+" 50: "+str(no_of_50s)+" 10:
"+str(no_of_10s)+" 1: "+str(no_of_1s))
```